

- 4 A tractor of mass 1000 kg pulls a trailer of mass 1000 kg via a tow-bar. The total resistance to motion has a constant value of 4000 N. One fifth of this resistance acts on the trailer. If the maximum breaking force of the tow-bar is 3000N, what is maximum acceleration of the tractor and trailer?

A 2.2 m s^{-2}

B 3.0 m s^{-2}

C 3.8 m s^{-2}

D 7.0 m s^{-2}

Ans: A

For the trailer, $\Sigma F = ma$

Max breaking force – 800 = 1000 a

$$a = (3000 - 800) / 1000 = 2.2 \text{ m s}^{-2}$$

- 5 A cylinder of radius r made of material A and B is placed on a rough floor as shown. The portion